

Chapter 10

Metal Surface Chemical Composition and Morphology

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10.1 INTRODUCTION

In microelectronic fabrication, metals are used as electrically conducting materials for interconnects. The properties such as resistivity, lithography process compatibility, melting point, thermal expansion coefficient, diffusivity in an insulator, adhesion to the insulator, and interatomic distances (stress) of the metal must be considered when selecting an appropriate interconnect material. The metal interconnect undergoes various processes during the fabrication of the integrated circuit (IC) such as etching, deposition, removal, and cleaning, and must withstand severe process conditions. Furthermore, the speed of the device is also attributed to metal properties.

Considering these many criteria, aluminum (Al) was chosen as an interconnect material and was widely used in early devices [1]. However, as the demand for miniaturization and speed of devices increased, Al was replaced by copper (Cu), for some device types, because of its low resistivity and high electromigration resistance [2]. The resistivity of the material plays a significant role in achieving higher speeds for the IC devices because resistivity–conductivity delay is strongly associated with the electrical and physical properties of the material [3]. The effect is more pronounced when the feature size is less than 100 nm. The resistivity of interconnect metals is provided in Table 10.1-1 [4]. Tungsten (W) is also used as a contact in both Al and Cu interconnect technology because it is easy to process. Even though Cu is currently used as interconnects, the search for suitable material for future interconnects is still of great interest. This is because, when feature size is below 10 nm, the resistivity of the copper increases because of the increase in grain boundary as per the Mayadas–Shatzkes model [5]:

$$\rho = (\rho_o + \rho_i) \left[1 + \frac{3}{8} (1 - p) \lambda \left(\frac{1}{t} + \frac{1}{w} \right) \cdot s + \frac{3}{2} \left(\frac{R}{1 - R} \right) \frac{\lambda}{G} \right] \quad (10.1-1)$$

where ρ is the total resistivity, ρ_o is the bulk resistivity, ρ_i is the impurity resistivity, p is the surface scattering coefficient, s is the roughness parameter, t is the line thickness, w is the linewidth, G is the average grain size, and R is the grain boundary reflection coefficient.

The thickness of the metal film in comparison with the mean free path of the electrons (λ) and surface roughness of the film are attributed to the change in total resistivity. Thus materials such as Al, W, and cobalt (Co) have been under consideration for future interconnect materials because they exhibit lower resistivity at 10 nm linewidth [6]. Fig. 10.1-1 shows the resistivity of Cu versus W and Co as a function of linewidth. Cobalt is also used as a barrier

TABLE 10.1-1 Properties of Interconnect Materials

Material	Thin-Film Resistivity ($\mu\Omega\text{-cm}$)	Melting Point ($^{\circ}\text{C}$)
Cu	1.7–2.0	1084
Al	2.7–3.0	660
W	5–15	3410
Co	5.6–9	1495
TiN	50–150	2950

Compiled from data in L. Taylor, Metals Handbook, Properties and Selection of Metals, Am. Soc. for Metals, eighth ed., vol. 1, 1961.

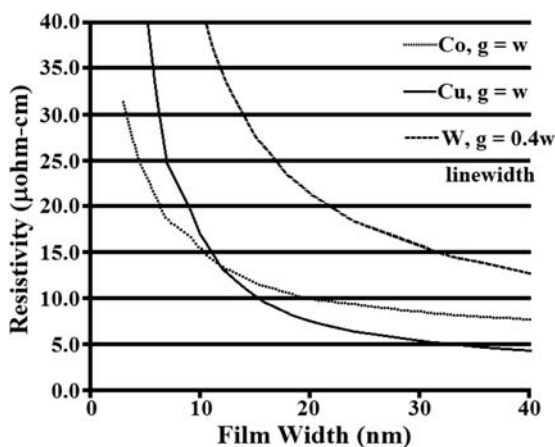


FIGURE 10.1-1 Resistivity of Co, Cu, and W as a function of film linewidth. Here “g” represents the average grain size divided by metal width and “w” represents the width of metal [9]. Used with permission of SEMI.

metal for Cu interconnects; it can control the diffusion of Cu atoms into dielectric materials at the sub-10-nm technology node [7,8].

Since metals are prone to oxidation because of high surface energy, passivation of metals is one of the critical steps in the semiconductor fabrication process. Passivation is indicative of a stable film formed on the metal surface that retards the surface reactivity with the surrounding environment [10]. A passivation layer is necessary for metal interconnects to prevent the metal from being oxidized and defects from occurring at subsequent processing steps and during storage. The surface chemistry of the passivated surface plays a critical role in removing contamination from the metal surface during the cleaning process. Thus careful screening of passivating agent should be carried out during the cleaning process.

In the present chapter, four different metals, Al, Cu, W, and Co, that are commonly used for semiconductor manufacturing are considered. First, the surface morphology obtained using various deposition techniques is discussed. Then, the passivation process and the surface chemistry of passivated metal surfaces are presented. Finally, the cleaning processes employed in the semiconductor industry to remove the contamination on passivated surfaces is reviewed.

10.2 ALUMINUM

10.2.1 Surface Morphology and Composition of Aluminum

Metallization is one of the final fabrication steps used for the interconnection of ICs. At the forefront of the semiconductor manufacturing era, Al emerged

as the most commonly used interconnected material for the metallization of IC, discrete diodes, and transistors devices [11]. As shown in Table 10.1-1, Al is one of the most conductive elements, and it has a resistivity of 2.7–3.0 $\mu\Omega\text{-cm}$, whereas Cu has a slightly lower resistivity of 1.7–2.0 $\mu\Omega\text{-cm}$. However, Al can be more easily dry etched than Cu to form interconnecting lines. Typical Al film thickness of $\sim 1\text{ }\mu\text{m}$ and conductor widths of 2–25 μm were commonly used in the early years of IC interconnects, whereas current thickness can be as thin as 200 nm and linewidths can be narrow to 10 nm. Aluminum is a polycrystalline material and offers many advantages over other potential interconnect metals, such as low cost, lightweight, high conductivity, ease of deposition and etching, good adherence to both silicon and silicon dioxide, and ability to form ohmic contact to p-type silicon (Si) and highly doped n-type Si [11–13]. Owing these advantages, Al is widely being used as interconnects in integrated devices in cell phones, tablets, and computers. Moreover, it does not contaminate the Si surface since it forms an Al_2O_3 (aluminum oxide) protective layer.

In aluminum metallization, the deposition process can be classified into two typical methods; chemical vapor deposition (CVD) and physical vapor deposition (PVD). The basic PVD methods are evaporation and sputtering; PVD by sputtering is the most commonly used method for depositing thin-film Al in the microelectronic industry. Both evaporation and sputtering processes have three steps: (1) converting the condensed liquid or solid phase into a vapor phase, (2) transporting this gaseous phase from the source to the substrate, and (3) condensing the gaseous source on the substrate [14,15].

Deposited Al films are commonly polycrystalline in nature irrespective of the deposition techniques and consist of many small monocrystalline grains that are the main cause for electromigration in Al lines [16]. However, the addition of a small weight percentage of Cu (0.5–4 wt%) to Al can significantly improve its electromigration resistance. In addition, Al interdiffuses into Si to form Al spikes. This problem can be solved by the addition of 1 wt% Si to Al to form an alloy [17,18]. Fig. 10.2-2 shows the surface morphology of Al with different concentrations of Si [19].

The microstructure of Al thin films has a direct impact on the performance of the IC devices. The electromigration behavior and mean time to failure are strongly dependent on grain boundaries, grain size, and crystal orientation. The morphology of bare Al is in the form of coarse columnar structure [19]. Usually, Al is stable when the texture is along the (111) preferred orientation, which can be obtained by various deposition techniques. This orientation of plane indicates the maximum surface density of Al atoms [20]. However, grain size and grain boundary are dependent on the type of deposition method and condition. Pilippenko et al. [21] reported changes in the structure of Al films on Si under both prolonged and rapid thermal treatments. The data are based on the Monte Carlo method and describe the change in the size and angles at triple points of the Al grains depending on the temperature and time of the

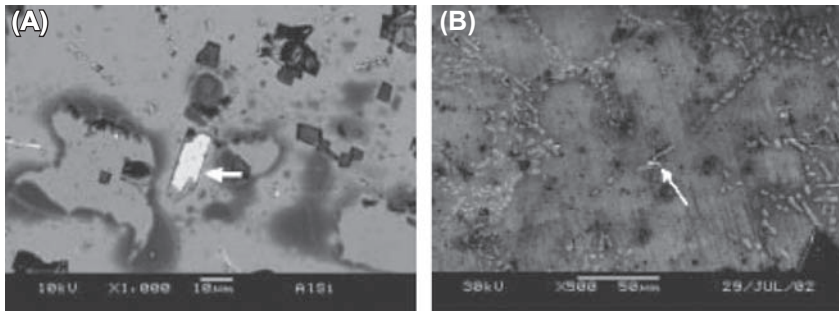


FIGURE 10.2-2 Scanning electron microscopy images of Al-based alloy with (A) 3.0% silicon and (B) 7.0% silicon [19]. Reprinted with permission from Z. Liu, M. Wang, Y. Weng, T. Song, Y. Huo, J. Xie, *Materials Transactions* 44(10) (2003) 2157–2162. Copyright ©, The Japan Institute of Metals and Materials.

thermal treatment. Larger grain sizes (and larger triple point angles) can lead to more stable metal films.

To obtain controlled film thickness, physical conditions, such as temperature and pressure, with respect to the sputtering process must be carefully controlled. Hence, sputtering techniques are widely used to deposit Al films with a uniform thickness over large surface areas. The sputtering process has advantages such as the ability to deposit refractory metals at high temperature, the capability to deposit alloy compositions, and the ability to deposit metals at a high rate. Loo et al. [22] introduced a multistep sputtering deposition process to achieve uniform Al film on SiO₂. The desired step coverage (120–275 nm height) and critical thickness (10 nm) of the film on SiO₂ are achieved at high temperatures (400–500°C) with a deposition rate of 1.8 nm/s, which yields a stable Al film with grains of (111) orientation. Kolar et al. [23] have reported the deposition of Al thin films on Si using various PVD techniques, such as ion beam deposition (IBD), magnetron sputter deposition (MS), and thermal evaporation (TE). The surface morphology of the MS samples exhibited higher roughness than IBD and TE. The larger grain (1.5–2.5 μm) morphology of Al with high roughness is more resistant to electromigration-induced failures than small grain morphologies [24]. In another invention [25], the sputtering deposition of Al films using hydrogen (H₂) and argon (Ar) gas reduces the formation of hillock growth morphology.

However, the conventional PVD metallization techniques cannot be applied to filling high-aspect ratio trench structures because of insufficient step coverage and poor hole-filling characteristics, even though the films exhibit high conductivity [26–28]. Alternatively, CVD receives considerable attention because of its excellent conformal coverage, high throughput, and relatively low process temperature. Typically the Al film can be deposited using CVD by three methods: atmospheric pressure CVD, low-pressure CVD (LPCVD), and plasma-enhanced CVD. The Al films deposited by LPCVD

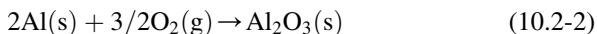
from triethyl aluminum on Si substrates show a smoother surface than Al films deposited by PVD at high temperatures [15]. However, LPCVD is an incompatible process for back-end-of-line deposition because it requires high deposition temperatures that are detrimental to IC devices. Thomann et al. [29] reported that the morphology of Al films deposited on Si by metal-organic CVD (MOCVD) shows that both the bottom and side conformity coefficients were not changed for structures with high aspect ratios.

Chemical vapor deposited Al films are unsatisfactory for semiconductor device manufacturing because of their high surface roughness, which deteriorates the surface morphology [30,31]. Therefore, atomic layer deposition (ALD) processes were invented to grow thin films with good step coverage. There has been very limited work done on the Al ALD processes; however, one such effort resulted in good conformal coverage and suitable resistivity of $\sim 3.4 \mu\Omega/\text{cm}$ [32]. This process involves a sequential series of gas-surface reaction under surface saturation conditions. Precursor and oxidizer are pulsed, separated by inert gas purging periods to prevent gas-phase reactions to obtain self-limiting film deposition [33]. The major difference between ALD and CVD is the Al layer thickness is controlled in the MOCVD process by the partial pressure of the reactant gasses and deposition time, whereas the Al layer thickness is controlled in the ALD process by the number of cycles [34]. Katamreddy et al. [35] reported the ALD 5.2-nm Al_2O_3 films obtained are amorphous in nature without any silicate formed at the interface. The researchers also deposited denser homogeneous Al films using electroplating directly on undoped P-type Si substrate with a high resistivity of $\sim 10^{-6} \Omega\text{-cm}$, whereas Al deposited directly on n-type Si yielded a resistivity between 0.3 and 1.0 $\Omega\text{-cm}$ [36]. Aluminum ALD has not yet been used industrially for ICs; further advancement of ALD process and its chemistry is required to deposit uniform thickness and excellent conformal layers on Si and SiO_2 .

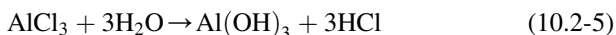
10.2.2 Aluminum Passivation

Aluminum naturally grows a passivating oxide (shown in Eq. 10.2-2) when exposed to air. This layer protects against corrosion because pure Al is thermodynamically unstable because of its high Gibbs free energy [37,38]. Since Al_2O_3 is thermodynamically stable, it can be used as a surface passivating film (due to less reactivity). However, it corrodes when exposed to aggressive aqueous environments, such as chloride (Cl^-), sulfate (SO_4^{2-}), nitrate (NO_3^-), or fluoride (F^-) ions [39]. Corrosion is defined as the deterioration of materials by electrochemical and chemical processes. The dissolution of Al takes place by redox reactions; i.e., the oxidation reaction occurs at the anode and the reduction reaction occurs at the cathode. The anode and cathode need not be two separate electrodes; for example, the n-channel and the p-channel are two separate electrodes, but can be two different sites in the same metal. When the Al is exposed to a high-pH solution, initially, the metal atom at the surface

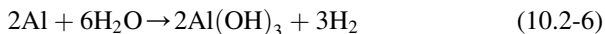
dissolves into the alkaline aqueous solution, leaving a negative charge on the surface metal ion. Secondly, corrosion continues as a depolarizer, such as oxygen (O_2), removes electrons from the metal.



Chlorine plays a major role in the corrosion of Al through the following reaction:



Once HCl is formed from the reactions in Eqs. 10.2-3–10.2-5, no additional Cl^- is required. The chloride ion only acts to facilitate the corrosion of the Al surface, as shown in Eq. (10.2-6). Hence, the presence of a small amount of Cl^- will cause severe local corrosion of the Al lines as the Cl ions recycle during the corrosion process, resulting in pitting [28].



The most common corrosion mechanism for Al interconnecting lines is pitting corrosion during the wet process [18]. Aluminum is covered with a thin oxide film, which inhibits anodic dissolution. However, the pits on the Al surface occur where weak oxide sites present themselves in the metal. Pits propagation initiates as the oxide surface adsorbs the chloride ions, then these ions penetrate through the oxide film, and finally, the Cl ion dissolution occurs at the Al– Al_2O_3 film interface [13,18,40]. Similarly, when the Cu content in the Al is high, a local cell exists between Al and Cu, which favors Al oxidation at the Al surface and water reduction at the Cu surface that eventually leads to corrosion [41]. This kind of corrosion is further enhanced when Al–Cu–Si line is connected externally (the absorption of moisture on the Si surface acts as a connector) to the p^+ -n junction of Si through a contact hole, for example. This is because when the junction is irradiated with light, photoelectromotive force moves the holes from the p^+ junction to the Al–Cu–Si side. This phenomenon causes the electrons to drift from the Al–Cu–Si line to the p^+ -Si substrate, which facilitates the Al oxidation reaction at the surface of Al [41].

A strategic method to prevent both the cathodic and anodic redox electron exchange from taking place is by coating the Al with organic and/or inorganic chemicals [17]. Very few literature references are available that discuss the protection of Al as an interconnecting material with organic or inorganic chemicals because of the presence of oxide layers [17,18,38,39]. Stupian et al. [39] proposed four criteria for substrate protection by a surface oxide: (1) low electronic conductivity; (2) low ionic conductivity; (3) low solubility; and (4) proper coordination with the substrate metal.

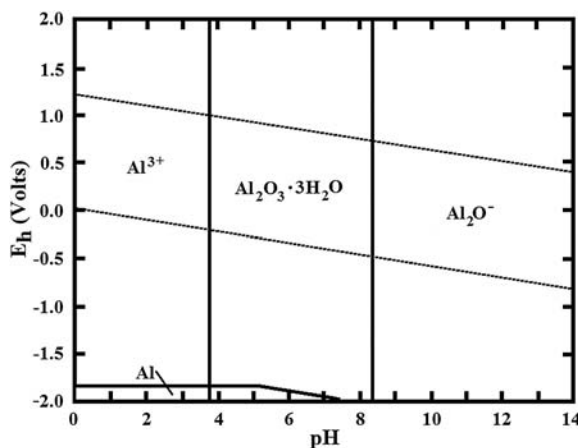


FIGURE 10.2-3 Pourbaix diagram for the Al–H₂O system at 25°C. Dotted lines are water stability lines [40]. Copyright National Association of Corrosion Engineers (NACE) International 1974. Used with permission.

The Pourbaix diagram of Al–H₂O system, shown in Fig. 10.2-3, contains pH regions where various species, Al (solid), Al₂O₃ (solid), Al³⁺ ions, and AlO₂[−] ions, are stable [40]. The corrosion region is identified where dissolved ionic species are stable: ions such as Al³⁺ at low pH, <4, and AlO₂[−] at high pH, >8. The passivity region is where a stable protecting oxide film (Al₂O₃) is formed, either a solid oxide or a solid hydroxide, between pH 4 and 8. It is to be noted that both corrosion and passivation occurs via oxidation reaction. However, for corrosion, the material removal occurs via dissolution of metal ions, whereas for passivation, a stable oxide layer forms on the metal surface that prevents the redox reaction from occurring between the metal and the electrolyte solution. The immunity region is identified when the stable species is in the form of an unreacted metal species, under reducing conditions [42]. An explanation of Pourbaix diagrams is given in Chapter 2.

Aluminum forms duplex passivation layers with an inner layer of Al₂O₃ and an outer layer of Al(OH)_x, based on quartz crystal microbalance, electrochemistry, X-ray photoelectron spectroscopy (XPS), secondary ion mass spectrometry, and Auger electron spectroscopy surface data [43,44]. The properties of the Al₂O₃ formed on the Al surface depend on various factors such as temperature, electrolyte composition, and distribution of microdefects, such as vacancies in the oxide film. The native oxide layer formed on the Al surface when it is exposed to air is amorphous in nature. The thin layer adjacent to the metal interface acts as a barrier, whereas the outer layer is more porous in nature. The oxide layer is crystalline and porous in nature when it is grown from sulfuric acid or oxalic acid solutions. The temperature plays a significant role when the oxide layer is grown by thermal oxidation. The

thickness of the Al_2O_3 layer is usually 1 nm, when formed in a dry O_2 environment. However, the thickness of the oxide layer increases upon heat treatment and in moist environments [45]. These Al_2O_3 films are quite resistant to corrosion, as they are insoluble and exhibit a semiconductor property except under very acidic or caustic or other special conditions (Cl^- (aq) exposure). Compared with bulk Al_2O_3 whose band gap is about 8–9 eV, the Al_2O_3 passive layer formed on the Al surface has a band gap of 3–4 eV. The electronic properties of the oxide layer under various conditions have been reported in the literature [46]. Corrosion of Al also occurs, as opposed to the pitting-type corrosion, when Al_2O_3 layer undergoes hydrolysis in the presence of hot water and, forms AlOOH and an $\text{Al}(\text{OH})_x$ layer at the top of the Al_2O_3 layer. The Al_2O_3 acts as a passivating layer because of the limited diffusion of oxygen or hydroxyl ions through this native oxide. However, the diffusion of these species is higher for the hydrated oxide layer and eventually results in uniform corrosion [47].

10.2.3 Aluminum Corrosion Caused by Processing

In IC production, the Al interconnect is usually capped with a TiN antireflective layer and with an underlying Ti/TiN barrier layer. The Al etching is typically performed with a Cl-based gas chemistry in a plasma environment (shown in Fig. 10.2-4). At the conclusion of the etching process, an F-based gas may be introduced to exchange a portion of the Cl^- in the postetching residue [48,49], with the intention of reducing corrosive effects. The concentration of these halide ions within the residue film can influence the chemical stability of the Al metal line, especially when immersed in an aqueous solution, such as a cleaning bath. Fig. 10.2-5 shows the surface morphology of Al after corrosion [50].

The chlorine atoms, which are responsible for Al corrosion, are either bound to the Al or to the carbon that is contributed from the etching gas or the photoresist. To remove the chlorine from the Al, various plasma process chemistries, such as CF_4 , O_2 , or CF_4 , with O_2 have been investigated. It is found that CF_4 plasma rich in O_2 is the most effective in removing chlorine bound to both Al and carbon. The susceptibility to corrosion is also determined by the composition of the substrate. For example, it is observed that Ti-capped Al is more susceptible to corrosion than bulk Al. Mayumi et al. [51] showed that the most effective way of controlling chlorine concentration is the partial removal of photoresist from the Ti-capped Al features using O_2 plasma.

To protect the postetched Al surface from a corrosive environment, Baek et al. [50] treated it using SF_6 plasma at 300 mTorr. They also studied other fluoride-related passivation compounds and found that corrosion occurred after air exposure at the grain boundary rather than at the crystalline surface because of the presence of residual chlorine atoms. The SF_6 plasma treatment etches the Al surface, whereas the fluorine atoms form a passivation layer on it. This

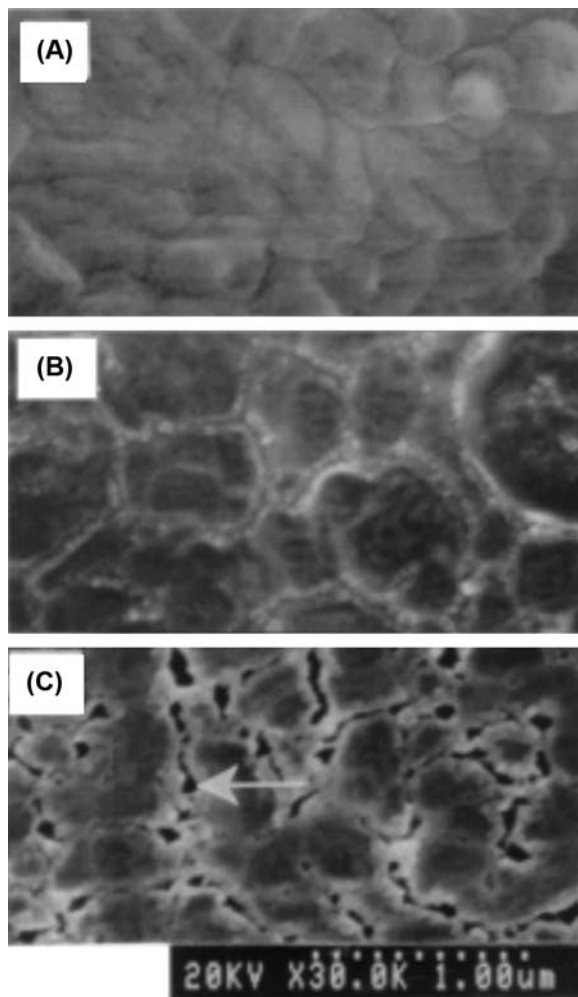


FIGURE 10.2-4 Scanning electron microscopy photographs of the Al–Cu alloy film surface: (A) as-deposited surface, (B) etched surface with the SF_6 plasma treatment in the pressure of 300 mTorr, and (C) etched surface without the SF_6 treatment [50]. Reprinted with permission from K.H. Baek, Y.S. Yoon, J.M. Park, K.H. Kwon, C.I. Kim, K.S. Nam, *ETRI Journal* 21(3) (1999) 16–21. Copyright © ETRI Journal, Korea.

fluorine passivation layer prevents the penetration of moisture onto the etched surface and suppresses the corrosion.

Both etchant gasses and wet etching chemistries can selectively attack the unprotected area of the Al thin film and produce undesirable by-products [52]. However, conventional wet-cleaning solutions, such as diluted HF (hydrofluoric acid) or NH_4OH , deteriorate the Al surface because of the presence of significant variations in the oxidation potential between the Al metal and the

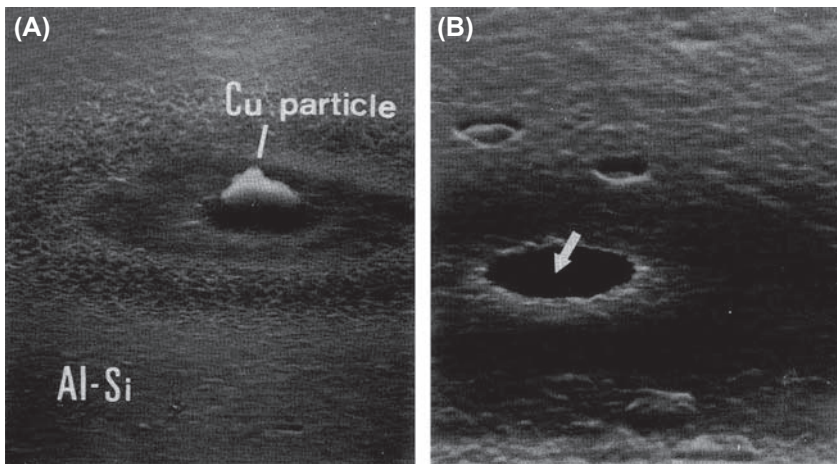


FIGURE 10.2-5 Scanning electron microscopy images of (A) Al-Si surface with Cu particles after immersion in deionized water (DIW) and (B) corrosion pit on Al-Si-Cu surface (0.5%) after immersion in DIW [41]. Reprinted with permission from N. Hayasaka, Y. Koga, K. Shimomura, Y. Yoshida, H. Okano, *Japanese Journal of Applied Physics*, vol. 30, 7R, 1991. Copyright 1991 The Japan Society of Applied Physics.

barrier metal [53]. In addition, deionized water (DIW) degrades the exposed Al surface by forming a thin native oxide layer and by removing aluminum atoms through diffusion, especially in high-resistivity DIW. Hence, corrosion occurs and degrades the interface between the metal and dielectric layers, which reduces the performance of the IC. A critical review on device failures due to Al corrosion in microelectronic fabrication industries is presented in the literature [54].

Extensive studies were carried out within the IC industry to develop a suitable chemical formulation that could remove the postetch residue from the Al surface and thereby control the corrosion. Chapter 4 discusses various cleaning processes for Al. Early work using amine-based chemistry by Lee et al. [55,56], to replace n-methyl-pyrrolidone, removed the photoresist and plasma etching residue by using hydroxylamine (NH_2OH), and an alkanolamine. More recent work was presented using a hydroxylamine, organic base, and acid in the pH range of 7–9 [57]. The use of a hydroxylamine along with amine-catechol corrosion inhibitor solution was proposed [58] to remove residues. In another early investigation, cleaning solutions that include H_2O , a hydroxylammonium salt, amines/quaternary ammonium hydroxides, and optionally a chelating agent and/or nonionic surfactant, wherein the pH is between 2 and 6, were used for removing plasma-etching residues [59].

After cleaning with an alkaline hydroxylamine solution, corrosion of Al is obtained when an intermediate DIW rinsing step is not carried out. Similarly, in another investigation, Andreas et al. [60] added citric acid as a corrosion

inhibitor, plus other additives, such as chelating agents, buffers, oxidants, antioxidants, and surfactants, to the hydroxylamine solution.

A widely used postetch residue remover class is semiaqueous chemistry (SAC) in which dilute sulfuric acid and hydrogen peroxide (DSP), with or without HF, is included [61,62]. In DSP solutions, H_2O_2 oxidizes the Al to Al_2O_3 , and this oxide is further etched and undercut by the sulfuric acid, thereby exposing Al. The purpose of adding H_2O_2 is to increase the oxidation rate of Al. Likewise, the addition of HF allows attack of the residues present on the SiO_2 surface via undercut mechanism [61,62]. Researchers [63] suggested an amine-type SAC solution that contains a corrosion inhibitor (gallic acid or catechol), water, organic amine, two or more water-miscible organic solvents, surfactant, and fluoride. They found that the resist reattachment and corrosion are reduced by improving the peelability of Al upon dilution with water by adjusting the composition of the solution.

Aluminum post—chemical mechanical polishing (CMP) cleaning processes, for example, for replacement metal gates [64] or in some cases Al damascene interconnects, have been studied [65]. The CMP processes leave a significant amount of slurry, other residue, and metallic contamination on the surface, which damages the surface layer. Therefore, post-CMP cleaning is necessary before the next deposition process. Chapter 5 discusses the post-CMP cleaning processes for Al. Very limited literature is available on post-CMP cleaning formulations for Al films. However, Yun et al. [66] introduced a novel post-CMP cleaning solution containing organic acid and chelating agent to reduce the galvanic corrosion behavior by minimizing the potential difference between Al and barrier metal. They obtained a resistance (R_s) for Al four times lower than that of W and two times higher than that of Cu for sub-100-nm device applications.

10.3 COPPER

10.3.1 Surface Morphology and Composition of Copper

Unlike Al, Cu interconnects are made using a damascene technique in which Cu is first deposited using electroplating into predefined features such as vias and trenches and then the excess Cu material is removed by CMP [67]. Because the diffusivity of Cu in SiO_2 is higher ($1.2 \times 10^{-11} \text{ cm}^2/\text{s}$ at 450°C) than that of Al ($1.65 \times 10^{-16} \text{ cm}^2/\text{s}$ at 1000°C), usually a thin barrier metal is deposited between the Cu and dielectric layers. To initiate the metallization process, a seed layer of Cu is deposited by PVD after the PVD or ALD barrier metal deposition, followed by electrochemical deposition (ECD) of bulk Cu onto the remaining gap in the features. The bulk Cu can be deposited by various methods, such as PVD, CVD, and electroless plating, in addition to ECD. However, ECD offers more advantages over PVD and CVD, which includes high-quality Cu deposit, conformal deposition in the trench and via,

and high throughput [68]. Also, the deposition rate is higher compared with electroless deposition. Electrochemically deposited Cu has high electromigration resistance and low electrical resistivity because larger grain size is obtained. Thus Cu ECD is widely used in microelectronic fabrication. Note that Cu deposited by ECD usually has an insignificant amount of impurities such as N (11 ppm), C (4.1 ppm), O (2 ppm), Cl (2.1 ppm), S (0.3 ppm), Ag (0.4 ppm), and others (<1 ppm) [69].

The deposits of Cu are polycrystalline with small grain size when using any of the discussed deposition methods. Thus, to increase the grain size to achieve lower resistivity, an annealing process is carried out, which also improves other characteristics of the deposit, such as stabilizing the microstructure of the Cu surface and reducing the number of grain dislocations (defects) [70,71].

The main requirement during ECD of Cu is superfilling; i.e., via and trenches are filled without any voids or seams because these defects affect the reliability of the devices. Thus, during electroplating, various additives are used to achieve superfilling [72,73]. These additives are categorized as levelers, brighteners, and accelerators. It is observed that during ECD, Cu grains nucleate from the bottom as well as the sidewalls of the trench [74]. Moreover, the seed deposit layer should be uniform to achieve superfilling. If the seed layer deposited by PVD has overhang or poor sidewall coverage, then it affects the ECD process uniformity. As 10-nm metal linewidths become more common, a uniformly deposited seed layer is necessary, which is not possible by PVD. Although the growth rate is lower for ALD processes, it can be a suitable option for smaller linewidths. Hence, ALD may be the appropriate choice for deposition of thin lines and deep vias. It is shown that a film thickness of 4.5 nm can be achieved using ALD processing [75]. Furthermore, the conformal deposition is possible even for high-aspect ratio structures. However, a proper choice of the precursor is needed to achieve conformal deposition [76–78]. The study shows that Cu film with texture <111> can be grown by this ALD method [79]. Researchers have found that the seed layer deposited by ionized metal plasma provides more uniform morphology and lower resistivity when compared with seed layer deposition by PVD [80].

The texture and microstructure of the Cu deposits are critical parameters, as they affect the electromigration resistance and resistivity of the Cu metal layer. For example, a large grain size is preferable because of its low resistivity, its bamboo grain structure, and its low electromigration resistance <111> orientation. As electromigration involves a mass flux of atoms along the grain boundaries, it can be controlled by reducing the grain boundary energy via an appropriate microstructure of the film. The texture and microstructures also affect the polishing and cleaning performance in subsequent CMP and cleaning process steps. The various ECD parameters such as temperature [81], pH [82], composition of the electrolyte bath [82–85], current density [86], current form (direct current or pulse plating) [87], agitation of the electrolyte [81], magnetic fields during deposition [88–90], substrate (Si/SiO₂/predeposited Cu/low-k)

on which deposition occurs [91,92], and reactor design [93] play a significant role in controlling the microstructure of the ECD Cu.

In addition to plating parameters, the size of vias and trenches also affect the grain size. Especially for the post-ECD process, annealing is carried out to increase the grain growth and to achieve the required microstructure [71,93–97]. In general, the grain growth is affected as the size of the trenches and vias is reduced. However, the ECD process is mature, and researchers expect that ECD will be employed with suitable modifications up to a feature size of 22 nm [82].

Typically, Cu ECD produces a significant number of defects, such as a high density of vacancies with small grain size and high grain boundary area along with inclusions from the electrolyte [98]. Fig. 10.3-6 shows the morphology of copper interconnect observed by scanning transmission electron microscopy [99]. The deposited films are in a thermodynamically metastable state because of the higher energy associated with the surface, grain boundaries, impurities, and strains. Thus as-deposited films tend to reduce their total energy associated with grain boundaries and other defects to attain a more stable state. This can be achieved by enlargement of grain size with a reduction in total grain boundary area, which is referred to as self-annealing [96]. Alternatively, annealing can be carried out in a furnace at optimum temperatures and time periods. During the annealing process, the microstructure of the deposited films stabilizes, which helps subsequent CMP processing by preventing via pull out, i.e., separation of via from the underlying metal line due to the formation of a void at the via dielectric–metal interface [100].

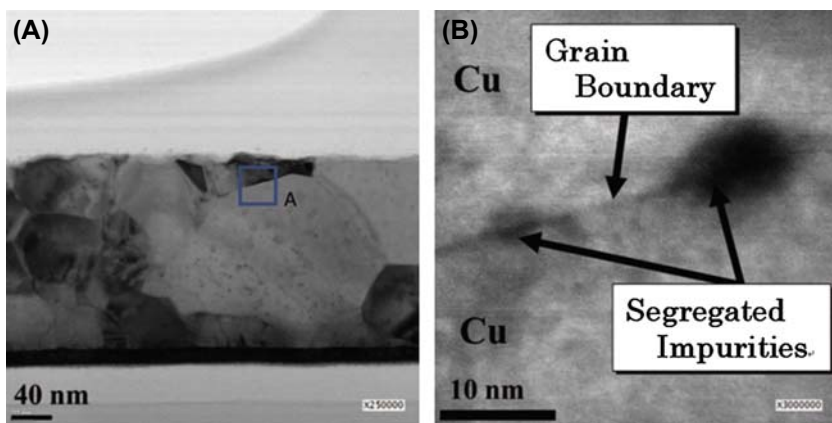


FIGURE 10.3-6 A cross-sectional aberration-corrected scanning transmission electron micrograph of (A) the structures of the Cu interconnect and (B) the dark-field image of region A in (A). The inserted arrows indicate grain boundaries and impurities [99]. Reproduced by permission of ECS—The Electrochemical Society from T. Nagano, K. Tamahashi, Y. Sasajima, J. Onuki, *ECS Electrochemical Letters*.

The effect of Cu microstructure and texture on CMP performance has been studied; uniform grain size and grain boundary structure (degree of misorientation) and grain orientation (texture) of $\langle 111 \rangle$ are commonly preferred to achieve a uniform surface roughness after CMP [101,102]. Uniaxed grain shows a lower surface roughness compared with twinned grains [101], and Cu texture $\langle 111 \rangle$ shows a lower removal rate because of the lower total energy associated with these films [101]. Also, pits generated after polishing are lower for Cu $\langle 111 \rangle$ grains [102]. The impurities added from the electrolyte bath chemicals into the grain boundaries also affect the defects formed during polishing [103]. The strain energy is at a minimum for Cu $\langle 200 \rangle$ texture and occurs from the mechanical stresses developed during the ECD deposition, which in turn affects the polishing performance. The Cu polish rate is higher for Cu $\langle 200 \rangle$ texture compared with $\langle 100 \rangle$ [102].

Although CVD and MOCVD have been considered for Cu deposition [73,103–109], the process is at the research level and has not been implemented in high-volume IC production.

10.3.2 Copper Passivation Processes

The following section focuses on Cu surface morphology because it strongly influences the passivation process. Copper passivation is necessary at various steps in microelectronic fabrication, such as metallization, CMP, and cleaning. Passivation controls the polishing rate and surface roughness during CMP and the surface composition during cleaning operations. Galvanic corrosion between Cu and barrier metals (Ta, Co, Ru) and stress-enhanced corrosion of Cu in contact with dielectric material due to the differences in thermal expansion coefficients are also experienced in semiconductor processing. The passivation layer should protect the underlying metal from diffusion of impurities and chemical reaction. Similarly, suitable chemicals are added to the CMP slurries and cleaning solution to modify the Cu surface.

In general, Cu forms a thin oxide film on the surface in aqueous solution or when it reacts with oxygen present in the air. The nature, stability, and protecting ability of the oxide depend on various factors, such as pH of the solution and the electrochemical potential developed at the metal–solution interface. The Cu–H₂O system Pourbaix diagram, shown in Fig. 10.3-7, reveals the thermodynamic stability of various oxide species at a given pH and the electrochemical potential [110].

The formation of a passivated film on Cu surface [111,112] occurs via a two-step mechanism:



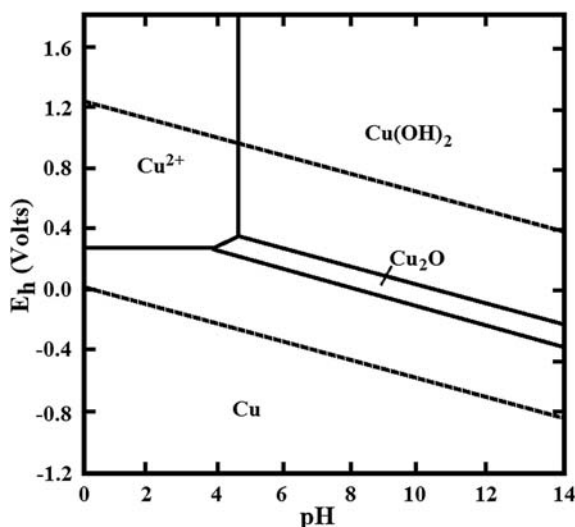


FIGURE 10.3-7 Pourbaix diagram for the Cu–H₂O system at 25°C [40,110]. Copyright National Association of Corrosion Engineers (NACE) International 1974. Used with permission.

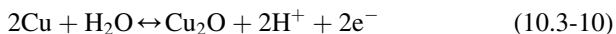
The Cu metal (M) is oxidized into metal (M^{n+}) hydroxide ions, which are adsorbed (ads) onto the surface. In a subsequent reaction, the adsorbed ions are condensed to form a metal oxide:



In a parallel reaction, the adsorbed ions dissolve in the aqueous solution, thus facilitating corrosion:



The Cu oxide consists of various forms of copper oxides: cuprous oxide (Cu_2O), cupric oxide (CuO), and copper hydroxide ($\text{Cu}(\text{OH})_2$). The ratio of these oxides in the passive film varies as a function of various ECD parameters, such as pH and electrochemical potential, which in turn depend on the composition of the electrolyte medium, temperature, and time. The composition of the passive films in various electrolyte media has been extensively studied using various analytical techniques [113–117]. The anodic reactions for the formation of oxides/hydroxides are given in Eqs. 10.3-10–10.3-12.



Typically, the passivation layer consists of an inner layer of Cu_2O and an outer layer of both CuO and $\text{Cu}(\text{OH})_2$. The thickness of the Cu_2O layer is about 1.2 nm and the thickness of the outer layer varies from 2 nm for CuO to 4 nm for $\text{Cu}(\text{OH})_2$ [118]. The stability of these layers varies as a function of pH and potential as observed in the Pourbaix diagram shown in Fig. 10.3-7. Both Cu_2O and CuO are stable over a wide pH range (from 4 to 13). However, with respect to potential, Cu_2O is less stable because it can be easily oxidized into thermodynamically more stable CuO at higher potentials. The formed oxides are either protective or nonprotective (if it is porous in nature) to the underlying metal depending on the prevailing conditions in the metal-electrolytic solution.

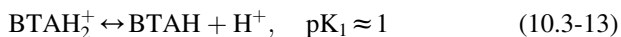
Feng et al. [113,114] studied the formation of oxide film on Cu surface using various electrochemical techniques, such as potentiodynamic polarization and electrochemical impedance spectroscopy (EIS), and other complementary techniques, such as XPS and scanning electron microscopy (SEM). The Cu surface is covered with a Cu_2O film when the $\text{pH} < 10$. However, the film is porous when the $\text{pH} < 3$, becoming more protective when $4 < \text{pH} < 9$ and passivation occurs at $\text{pH} 10$. At higher $\text{pH} (>12)$, a CuO film is formed. The nature of the passivated film also depends on aerated conditions, i.e., the concentration of dissolved oxygen present in the aqueous electrolyte. Research has shown that the formation of $\text{Cu}(\text{OH})_2$ film is feasible in aerated H_2O [114]. Also, the stability of the oxide film is dependent on the Cu texture. The Cu_2O film formed on a $\text{Cu} \langle 111 \rangle$ surface is more stable compared with that formed on other surfaces [115]. Owing to its porous nature, copper oxide films are more prone to having the underlying metal corrode compared with Al_2O_3 , as the enthalpy of the $\text{Cu}-\text{O}$ bond is lower (269 kJ/mol) than the enthalpy of the $\text{Al}-\text{O}$ bond (511 kJ/mol).

During both the Cu CMP process and the post-CMP cleaning process, a suitable corrosion inhibitor is added to the slurry, to control the surface quality thereby preventing dishing and erosion issues [119], and to the cleaning solution, to modify the surface to prevent corrosion or to scavenge dissolved O_2 that can lead to corrosion. Copper corrosion has been a critical issue ever since Cu became an interconnect material. However, the corrosion inhibitors based on Na, K, or Mo ions used in other industries (i.e., automotive, boiler) could not be employed in IC industry, as these ions are detrimental in semiconductor processing.

Various Cu passivation agents and corrosion inhibitors that can be used in IC manufacturing have been suggested, including tetrazole pi-electron-rich aromatics, triazole pi-electron-rich aromatics, imidazole pi-electron-rich systems, carbazole pi-electron-rich systems, N-based systems, and surfactants. Subramania et al. [120] compiled the various inhibitors proposed for Cu that are stable in acidic and neutral media. Most of the corrosion inhibitors are organic compounds containing heteroatoms, such as N, P, and S, which can form coordinate bonds through its free electron pair with the metal surface

[121,122]. Also, compounds with π -bonds have been proved to be excellent corrosion inhibitors, as they form a complex film on the metal surface via π -orbital bonding [122]. These inhibitors can form either a thin film or bulk precipitates on the Cu surface to retard the corrosion reaction, for example, during the polishing process, where the Cu is in contact with the slurry, which consists of small abrasive particles and various chemical agents that can initiate or accelerate corrosion. However, if the passivated surface is hydrophobic, it leads to particle contamination issues.

Among these various corrosion inhibitors, benzotriazole (BTA) and its derivatives are commonly used, especially in Cu post-CMP processes, and extensively studied [123–140]. The structure of BTAH is given in Fig. 10.3-8. BTAH can exist in one of the three forms depending on the pH of the slurry, as shown in Eqs. 10.3-13 and 10.3-14; BTAH at neutral or weak alkaline media, BTAH_2^+ in strong acidic media, and BTA^- in strong alkaline media.



The Pourbaix diagram of the Cu-BTAH- H_2O system shows the type of species thermodynamically stable at a given pH and electrochemical potential, Fig. 10.3-9 [110,129]. This diagram shows that a stable Cu-BTA complex is formed in the pH range 4–10. However, the interaction of BTAH with Cu depends on various factors, such as pH, the composition of the electrolyte medium, nature of the Cu surface, grain orientation of the Cu surface, and hydrodynamic conditions (static or stirred). The BTAH can be simply adsorbed onto the Cu surface (physisorption) or form a Cu-BTAH polymer complex (chemisorption). Langmuir isotherms [141] and Frumkin isotherm [142] have been suggested to model the BTAH adsorption on Cu surfaces. Cho et al. [143] studied the interaction of BTA with three different Cu surfaces using various analytical techniques, including contact angle measurements, EIS, and XPS. The results suggested that the interaction of BTA is strongly influenced by the presence of an oxide film on the Cu surface. The BTA film is thicker (2.9 nm) when there is no native oxide on the Cu. Various BTA derivatives, such as s-methyl-BTAH and 5-chloro-BTAH, also have been suggested as corrosion inhibitors [144]. It is found that the corrosion inhibition efficiency is strongly dependent on the side chain groups [139,145–157].

Although BTA is capable of forming a passivating film on Cu in acidic, neutral, and basic media, it poses a few issues such as slow inhibition, being a

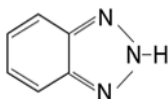


FIGURE 10.3-8 Structure of benzotriazole.

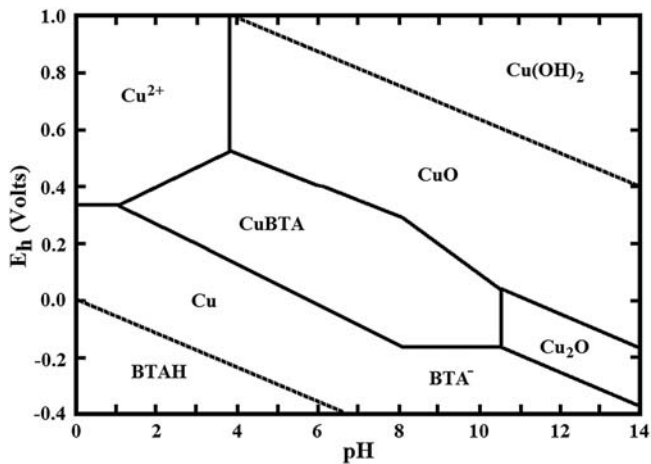


FIGURE 10.3-9 Pourbaix diagram of Cu-BTA-H₂O system. Adapted from *Nanofabrication*, M. Stepanova, S. Dew (Eds.), *Chemical Mechanical Polish for Nanotechnology*, 2012. Springer, Vienna, Austria, p. 239 chapter written by L. Nolan, K. Cadien. With permission of Springer.

nonenvironmental friendly chemical, and contributing to post-CMP challenges such as organic contamination. Thus finding a suitable corrosion inhibitor for Cu CMP is of great interest in the semiconductor industry. Uric acid [119], tetra (hydroxyethyl) ethylenediamine chelating agent [151], imidazole [152], and potassium sorbate [153] are a few other corrosion inhibitors proposed for Cu. Kim et al. [154] reported the inhibition effect of methyl-benzotriazole (MBTA) on the Cu CMP process. Their study confirms that MBTA exhibits better corrosion inhibition efficiency than BTA because of the formation of strong covalent bonds between methyl groups and the Cu surface. However, contact angle, EIS, and field effect SEM analysis show that the Cu-BTA complex is more easily removed than the Cu-MBTA complex during post-Cu CMP because of the presence of the methyl group in MBTA.

Liu et al. [155] have proposed a macromolecular organic compound that can form complex with copper oxides only when kinetic and frictional energy are imposed on the surface.

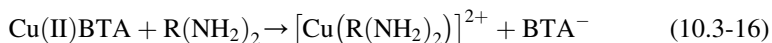
Lee and Robert [158] used oxalic acid dehydrate and an amine-based cleaning composition to remove the residual inorganic materials formed during the plasma etching of metal or oxide layers with halogen-containing gasses.

10.3.3 Contamination of Cu Passivated Surfaces

The passivated post-CMP copper surface, when covered with a BTA film, is hydrophobic in nature. This BTA film poses severe challenges, including issues during wafer drying and poor adhesion of subsequent layers deposited

on the copper surface. After CMP, a cleaning step is required to remove this passivated layer using appropriate cleaning solutions, which leaves the Cu surface passivated but hydrophilic [159,160]. Furthermore, the slurry particles used in CMP processing are attracted to the Cu surface [156,157,161]. These particles attached to the Cu surface are removed via both chemical and physical forces. The basic mechanism is to reduce the electrostatic attraction between the particle and the Cu surface via suitable cleaning chemistry and by applying proper physical force (e.g., megasonics, brush scrubbing) to remove the particles from the surface without redeposition. The main criterion of the cleaning chemistry is to provide repulsive interaction forces by controlling the surface zeta potentials. Furthermore, the cleaning solution also should be able to remove organic residues and metallic contaminants if present without increasing the surface roughness plus passivate the Cu surface with a film hydrophilic in nature. Chapter 5 reviews the adhesion and removal mechanisms for post-CMP particle removal.

There are several reports on post-CMP cleaning formulations. HF-based cleaning formulations are not recommended for Cu because they induce corrosion and are detrimental to Si-based dielectrics. Murakami and Ishikawa [162] suggested that the pH of the cleaning solution should lie outside the range 3–10 because BTA forms a strong complex with Cu. However, Lin et al. [163] observed that the cleaning solution with pH 5 containing citric acid can remove organic contamination from Cu surfaces. This implies that the electrochemical potential at metal-electrolyte solution plays a critical role in removing organic contaminants. Furthermore, the addition of NH_4OH to citric acid facilitates increased particle removal [164]. However, later, Yeh et al. [157] found that the surface roughness after cleaning is much better in a solution containing buffered hydrofluoric acid (BHF) and aqueous ozone (O_3) than in a cleaning solution containing citric acid. Researches also focused on alkaline-based cleaning solutions for post-Cu CMP cleaning [165,166]. An alkaline solution having high pH values can remove only CuO and hence the Cu surface is passivated with Cu_2O [167]. Also, higher particle removal efficiency can be achieved in alkaline solutions, as the particles can acquire a large negative charge. Among various alkaline solutions, tetramethyl ammonium hydroxide (TMAH) has been considered as a potential candidate for post-Cu cleaning [159]. Also, to remove BTA and surfactants from the Cu surface during cleaning, an amine-type chelating agent is suggested by researchers [168]. The structure of the cleaning agent is given in Fig. 10.3-10. This chelating agent dissolves the Cu-BTA complex through the chemical reactions shown in Eqs. 10.3-15 and 10.3-16.



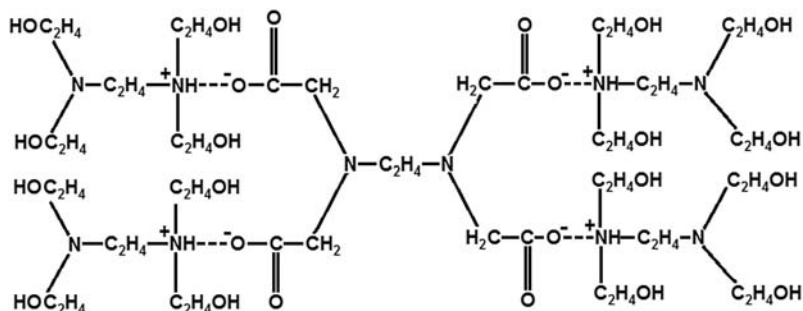


FIGURE 10.3-10 Structure diagram of the chelating agent used to remove benzotriazole from copper surfaces [168]. Reprinted with permission from Z. Gu, Y. Liu, B. Gao, C. Wang, H. Deng, *Journal of Semiconductors* 36 (10) (2015) 106001. A novel compound cleaning solution for benzotriazole removal after copper CMP. IOP publish this journal on behalf of the Chinese Institute of Electronics & the Institute of Semiconductors of the Chinese Academy of Sciences.

A surfactant is incorporated in the cleaning solution to prevent the Cu corrosion via adsorption on the Cu surface as shown in Fig. 10.3-11. Ramachandran et al. [160] suggested a hybrid cleaning technology for cleaning Cu/low-k materials during the post-CMP process. Non-amine-based cleaning solutions are also of keen interest; an amine such as TMAH has a low vapor pressure, but the molecule can pose severe health issues [169].

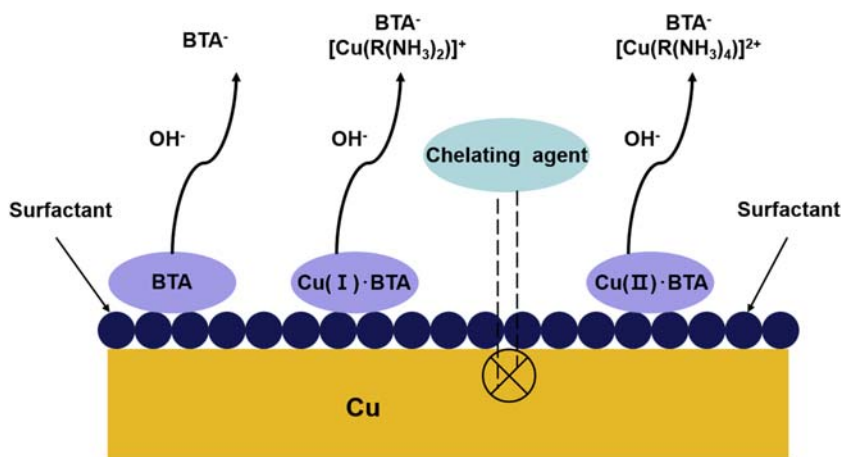


FIGURE 10.3-11 Schematic of benzotriazole (BTA) and Cu-BTA removal during post-chemical mechanical polishing cleaning. The schematic shows that the chelating agent reacts with the copper-complexed BTA, and not the copper or the BTA [168]. Reprinted with permission from Z. Gu, Y. Liu, B. Gao, C. Wang, H. Deng, *Journal of Semiconductors* 36 (10) (2015) 106001. A novel compound cleaning solution for benzotriazole removal after copper CMP. IOP publish this journal on behalf of the Chinese Institute of Electronics & the Institute of Semiconductors of the Chinese Academy of Sciences.

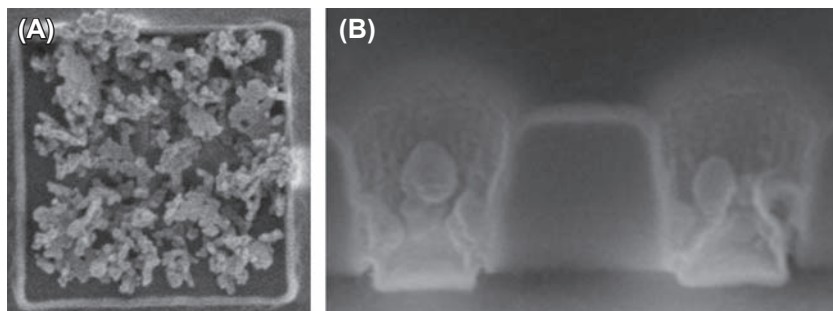


FIGURE 10.3-12 Scanning electron microscopy observation of Cu dendrite: (A) top image and (B) vertical image [175]. Reprinted with permission from Trans Tech Publication. From J.S. Heo, J.H. Oh, H.J. Shin, N.I. Lee, *Solid State Phenom.* 145 and 146 (2009) 331.

For Cu/low-k dual damascene processes, wet cleaning is employed to remove the polymeric residuals from the trench and via side walls after the etching process [170,171]. The etching process generates polymeric residuals that also are attached to the Cu surface. Also, resputtering of the Cu onto the side walls of the trench and via occurs. The polymeric residue on the Cu surface mainly consists of oxides and fluorides of copper and fluorocarbons [159,161]. Thus an effective process is required to clean the postetch residues on the Cu and the via and trench side walls without corroding the Cu as well as to remove the resputtered Cu before depositing barrier metal [170–172].

Cleaning solutions containing fluorine and amine compounds are commonly employed to clean the post-etching surface [170–173]. Tsang et al. [170] showed that cleaning the surface in a single wafer system using a fluoride-based cleaning solution is more effective compared with that of amine-based cleaning solutions in terms of electrical and reliability performance. The efficiency of fluoride-based cleaning solutions is also reported by other researchers [171,174]. Dilute HF solution is also suggested to remove the postetch residues [170,175]. However, during the cleaning process, corrosion and redeposition of copper can occur and lead to dendrite structure formation as shown in Fig. 10.3-12 [175]. To enhance the efficiency of removal of postetch residue removal, the use of megasonic wet cleaning is suggested [176]. Considering the undesirable environmental impact of fluoride-based cleaning solutions, non-fluoride-based cleaning solutions are of great interest. For example, a mixture containing choline chloride and malonic acid has been suggested to remove the postetch residues from the Cu surface [170].

10.4 TUNGSTEN

10.4.1 Surface Morphology and Composition of Tungsten

As discussed in Section 10.2, Al is a widely used material for the metallization of ICs, as it possess high conductivity and low contact resistance besides good

adherence to Si and SiO₂ [177,178]. However, the electromigration and spiking of Al through Si under high current densities, plus corrosion due to moisture or air exposure can seriously affect the reliability of the metallized circuits. Also, as dimensions of the microelectronic structures move toward smaller nodes, the use of tungsten (W) instead of Al [179,180] is preferred because of its better step coverage and excellent adhesion [180–182]. Tungsten is used as a diffusion barrier and deposited by various methods [183]. As shown in Table 10.1-1, W is a transition metal and has the highest melting point of all elements at 3422°C and a boiling point of 5660°C. The metallic form of tungsten can resist acids and alkalis [184]. It offers many advantages, including high conductivity, ease of depositing on Si and SiO₂ because of its flexibility (high tensile strength), its ability to withstand high temperatures under anaerobic conditions, its resistance to chemical attack by etching agents, and its use as an effective diffusion barrier for ohmic and Schottky contacts [179,185].

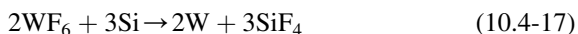
Many techniques have been reported for depositing W films, including low-pressure plasma spraying [186], CVD [187], thermionic vacuum arc method [188], sputtering [189], and laser ablation [190]. Among all these techniques, CVD processes are extremely attractive because of the low deposition temperature and the controlled morphology and composition of W films, which can be achieved by changing the deposition process conditions, such as pressure and temperature. CVD offers several advantages over other deposition methods, such as cathodic sputtering or evaporation, yielding high purity, uniform thickness, conformal step coverage, and high throughput [191].

Several authors [192–194] have reported the deposition of W using LPCVD to obtain conformal and good fill characteristics up to the 90-nm technology node. Kaplan et al. [195] investigated W deposition from tungsten hexacarbonyl (W(CO)₆) using a hot-wall LPCVD reactor at 200–580°C with H₂ as the carrier gas and with the partial pressure varying from 5 to 50 mTorr. The W films deposited at a temperature of 500°C had high resistivity (200–500 μΩ-cm) because of the presence of carbon and oxygen impurities. However, the resistivity is reduced to two to five times the bulk value (5.6 μΩ-cm at 20°C) when the films are deposited at temperatures ≥500°C. The polycrystalline α-W with a (100) preferred orientation is observed when the film is deposited at 450°C. Diem et al. [196] deposited W films containing 10 and 10 atomic % (at %) C and O, respectively, using a hot-wall CVD reactor at 420°C and atmospheric pressure. Vacuum annealing of the films for 1 h at 850°C increased the purity to 96 at% W and lowered its resistivity to 6.6 ± 12.5 μΩ-cm.

Lai et al. [197] deposited the W films on Si (100) from W(CO)₆ by LPCVD in an ultrahigh vacuum-compatible reactor. The polycrystalline β-W films with a (211) orientation and resistivities of >1000 μΩ-cm are deposited at 375°C with elemental composition approximately 80 at% W, 15 at% C and 5 at% O. However, vacuum annealing at 900°C converts this metastable β-W to polycrystalline α-W with small randomly oriented grains and nanoscale

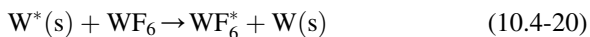
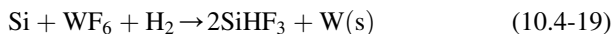
(100 nm) voids with a resistivity of $\sim 19 \mu\Omega\text{-cm}$. Films deposited at 540°C exhibited high-purity (~ 0.95 at%) polycrystalline $\alpha\text{-W}$, with low resistivities ($18 \pm 23 \mu\Omega\text{-cm}$) and a tendency toward a (100) orientation. However, vacuum annealing at 900°C further reduced the resistivity to $\sim 10 \mu\Omega\text{-cm}$ (approximately two times the bulk value) and resulted in a columnar growth morphology with (100) orientation.

Tungsten films deposited using WCl_6 on Si at atmospheric pressure in the temperature range of $525\text{--}625^\circ\text{C}$ are nonuniform with rough surfaces, whereas at a high pressure of 40 Torr in the temperature range $480\text{--}670^\circ\text{C}$ smooth films are obtained [198]. Typically, W films are deposited on Si wafers by LPCVD using WF_6 at 285°C [191]; the formation of W from WF_6 is shown in Eq. (10.4-17).



Before depositing W for the contact layer in Si layers (i.e., polysilicon gate, Si substrate), the dielectric contact hole is etched followed by the photoresist stripping step, which includes a cleaning step to remove the Si native oxide layer. It was found that the resistivity of tungsten layers decreased with increasing film thicknesses up to 150 nm. The formation of WSi_2 is achieved by annealing the W/Si structures at 650°C for 1 h [191]. This film can be used as a plug for DRAM (dynamic random access memory) deep trench or stacked DRAM vias owing to its lower resistivity [199]. In another investigation, Bain et al. [185] pointed out that the resistivity of the SiH_4/WF_6 layer decreases as the temperature increases above 450°C .

Bain et al. [179] and Melliar-Smith et al. [200] highlighted that H_2 -reduced CVD W films show excellent adhesion to Si substrates and also achieve good uniformity. The reduction reaction takes place:



where * denotes an activated surface site. Initially, the nucleation step takes place by the Si surface reacting with WF_6 to form a W seed layer. Next, this W seed layer initiates on the Si surface, which further helps the reduction of WF_6 by providing activated surface sites [201].

There is a need for decreasing the film thickness, especially for nanometer-scale devices in the semiconductor industry. Unfortunately, the conventional thin-film deposition techniques such as CVD and PVD fail to deposit nanometer-scale thin films. ALD is used to fabricate a variety of electronic devices and components. The ALD thin films are highly uniform, conformal, homogeneous, dense, and relatively free from defects. ALD uses gas–solid

chemical adsorption reactions over the surface. These chemisorption reactions are self-controlled and self-limiting, resulting in film growth by consecutive atomic layers [33].

Kalanyan et al. [202] used the ALD processes to deposit conformal and uniform 10-nm W films on SiO_2 substrates used for sub-10-nm high-storage memory devices. Other research confirmed that the addition of H_2 delays W nucleation on SiO_2 thereby expanding the ALD selectivity and reducing volcano defects [203]. In addition, fluorine-free [204] or low levels of fluorine W ALD [205] enables the continued scaling of three-dimensional NAND and DRAM devices. When fluoride is used in the W deposition process, it will diffuse into Si, which is detrimental to the performance of the devices. Nevertheless, ALD process modifications are needed, for example, changing the chemistry to deposit low-resistivity W with thinner nucleation, excellent conformality, and uniformity, plus less expensive processes that can help extend the technology to future nodes.

The deposited W films are polycrystalline in nature with body-centered cubic structure [206]. The crystal orientations of (110) and (114) are usually observed [206]. However, the (100) orientation is reported when the thickness of the W film increases because of longer reaction times [207]. Grain sizes varying from 14 to 80 nm are reported in the literature [206,207]. Fig. 10.4-13 shows the morphology of tungsten film with different grain sizes [206]. The morphology of the tungsten film depends on the deposition conditions, such as temperature and pressure. Gokce et al. reported [207] that the film is porous in nature when CVD processing is carried out at low temperature (285°C) and high pressure (10 Torr), whereas the film is continuous and smooth at high temperature (385°C) and low pressure (1 Torr).

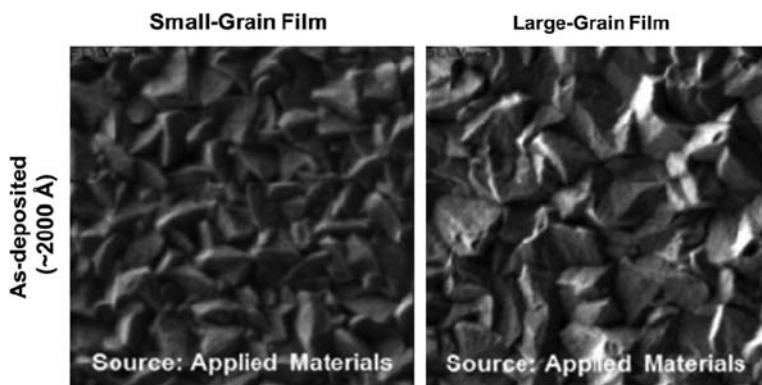


FIGURE 10.4-13 Scanning electron microscopy plan view of as-deposited chemical vapor deposition W films with different grain size, what is considered a small- and a long-grained film, using 1- μm field of view [206]. *Reproduced by permission of ECS—The Electrochemical Society from K. Xu, S.H. Shen, J. Fung, H. Iravani, I. Carlsson, T.Y. Liu, K. Mikhaylich, ECS Journal of Solid State Science and Technology.*

10.4.2 W Passivation Process and Contamination Issues

Passivation of W is necessary to protect its surface from aggressive environments during the fabrication of microelectronic devices. The typical W CMP process uses a single polishing step that not only removes the overburden materials formed on the wafer during metallization but also protects the surface from various aggressive environments by changing its surface composition, the formation of a thin passivating film [208]. These results depend on the slurry that is used during planarization, which typically contains a strong oxidizer. Bielman et al. [209] observed that no material loss occurs in the absence of an oxidizer during the W CMP process. In another investigation, the metal surface is passivated in a fraction of a second when it is exposed to the CMP slurry [210].

The role of oxidizer is crucial during the W CMP process because oxidants form a passivating film by converting tungsten to tungsten oxide. Tungsten is known to form various oxides; among them, WO_3 is the most stable, whereas WO and WO_2 are not considered stable protective oxides. Various oxidants have been reported for W CMP by several researchers, such as H_2O_2 [211], KIO_3 , $\text{Fe}(\text{NO}_3)_3$ [212,213], KMnO_4 [213], and KNO_3 [213]. The formation of oxide layers during the W CMP process is similar to that of Cu; WO_2 is believed to be present as a subsurface oxide below the outer WO_3 layer [211]. Tamboli et al. [183] used KIO_3 and H_2O_2 oxidizers in W CMP to passivate the surface. There is evidence that W shows an excellent passivation behavior in both KIO_3 and H_2O_2 solutions. However, the current density in the passivation regime for H_2O_2 is almost one order of magnitude higher than in KIO_3 solution. Tamboli et al. postulated the structure of the passive film has two layers; the inner compact barrier film is considered noncrystalline and glassy in nature and the outer precipitated layer results from cations emerging out of the barrier layer. The film contains three different bridges $-\text{H}_2\text{O}-\text{M}-\text{H}_2\text{O}-$, $-\text{HO}-\text{M}-\text{HO}-$, and $-\text{O}-\text{M}-\text{O}-$, where M is the metal W. During the oxidation process, the $-\text{OH}$ -type bonds are converted to $-\text{O}-\text{M}-\text{O}-$ type bonds, resulting in improved protective properties of the oxides. Basim et al. [208] reported that the protective oxide film is formed on the W surface with a thickness of 5–7 nm when H_2O_2 is used as an oxidizer, which enables an enhanced removal rate through the interaction between the nanoparticles in the slurry and the surface oxide hillocks.

Lim et al. [214] used three different types of oxidizers, namely, KNO_3 , KClO_3 , and KIO_3 , during W CMP. Among all, KIO_3 reveals a great extent of surface oxidation (8 atomic units thick). It also revealed that the surface roughness is reduced 20% in KIO_3 than in KNO_3 and KClO_3 . The researchers concluded that the polishing rate of W is enhanced because of surface oxidation and the formation of surface WO_3 species. Perry et al. [215] reported that, based on the oxidizing strength, $\text{NH}_4\text{OH} + \text{H}_2\text{O}_2$ exhibited the highest removal rate during W CMP compared with $\text{HCl} + \text{H}_2\text{O}_2$, $\text{KIO}_3 + \text{NaOH}$, and KOH .

In another investigation, the combination of mixed oxidizers is proposed, adding KIO_3 and $\text{Fe}(\text{NO}_3)_3$ into H_2O_2 for corrosion protection [216]. The researchers observed that the mixture of 0.1 M KIO_3 with 5 wt% H_2O_2 has a decrease in corrosion potential with an increase of pH, whereas 0.1 M KIO_3 with the mixture of 0.1 M $\text{Fe}(\text{NO}_3)_3$ and 5 wt% H_2O_2 has an increase in corrosion potential with a decrease of pH. Furthermore, the authors correlated their results with the Pourbaix diagram shown in Fig. 10.4-14. The KIO_3 -mixed slurry corrodes the W surface by forming WO_4^{2-} , whereas the $\text{Fe}(\text{NO}_3)_3$ -mixed slurry protects the W surface by forming WO_3 at low pH. Once WO_3 establishes, it prevents further oxidation and dissolution of the underlying W by the slurry. The main advantage of W CMP is that corrosion inhibitors are not required to protect the surface from aggressive environments compared with Cu CMP.

In another investigation, oxalic acid is used as a complexing agent and H_2O_2 and $\text{Fe}(\text{NO}_3)_3$ are used as oxidants for the W CMP process [217]. When both oxidants are used together, H_2O_2 is decomposed by a ferrous ion (Fe^{2+}) to produce hydroxyl radicals ($\cdot\text{OH}$) or by a ferric ion (Fe^{3+}) to produce perhydroxyl radical ($\cdot\text{OOH}$); the reactions are shown in Eqs. 10.4-22 and 10.4-23. However, the ferric ion is stable at higher pH, where it reacts with organics present in the slurry. The decomposition rate of H_2O_2 can be reduced by limiting the contact between H_2O_2 and Fe^{3+} [218]. Hence the authors stabilized Fe^{3+} by the adding a complexing agent to the slurry. The researchers also reported that the slurry containing $\text{Fe}(\text{NO}_3)_3$ produced a dense oxide layer on

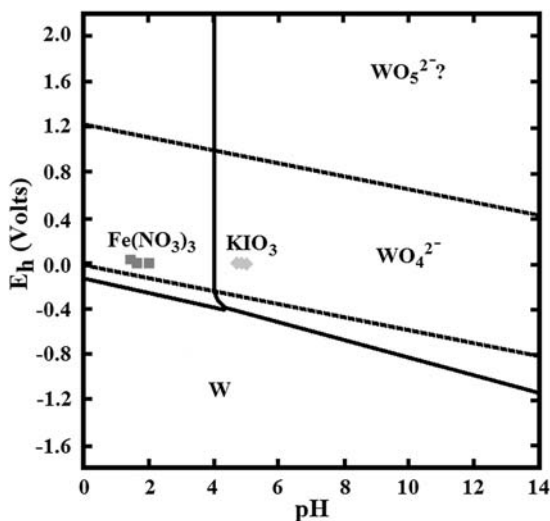
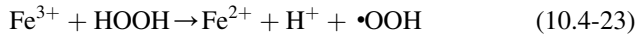
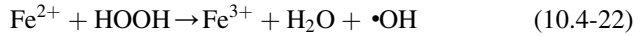


FIGURE 10.4-14 Pourbaix diagram for W-H₂O system at 25°C [40]. Copyright National Association of Corrosion Engineers (NACE) International 1974. Used with permission.

the W surface, whereas slurries containing oxalic acid and H_2O_2 induced a porous oxide layer.



Many kinds of defects are observed after CMP of the W film, which includes residual slurry particles, process-induced microscratches, erosion affected by pattern and pattern density, and corrosion by wafer stagnancy on the pad after CMP. Yan et al. [219] suggested the use of NH_4OH solution during brush scrubbing for post-W CMP instead of a HF solution. Both HF and NH_4OH etch and lift off particles during the W post-CMP brush scrubbing and effectively removes particles. However, HF chemistry used for W post-CMP cleaning degrades the electrical characteristics of a device [219]. Researchers observed that higher profiles achieve more effective brush contact and result in higher particle removal efficiency, when using NH_4OH compared with HF, without adding contamination. However, Jeong et al. [220] observed circular-shaped defects after post-W CMP cleaning. These defects are generated from the tungsten oxide ions dissolved in aqueous polishing solution during W CMP. However, they are removed by two methods, dipping post-W CMP wafer into a HF solution followed by additional oxide CMP, followed by polyvinyl alcohol (PVA) brush cleaning with a concentrated NH_4OH solution. The researchers reported that the use of concentrated NH_4OH solution during PVA brush scrubbing is a more efficient post-W CMP cleaning method compared with other methods. Chapter 5 outlines the various methods of post-CMP cleaning.

10.5 COBALT

10.5.1 Surface Morphology and Composition of Cobalt

Cobalt is a potential candidate for interconnects when the linewidth is <10 nm because it offers lower resistivity because of its short free mean path of electrons. Also, Co can be used in place of the Ta/TaN bilayer as a Cu diffusion barrier and can promote adhesion between Cu and a dielectric material for <32 -nm technology nodes [7,8]. The integration of Co or Co-based alloy materials requires compatibility with various wet process treatments such as precleaning, CMP, and post-CMP cleaning. During wet cleaning and CMP, Co is observed to exhibit high corrosion and dissolution rates. Although Co is a harder material compared with Cu, it is more susceptible to corrosion. Therefore, modifying the Co surface, with the intent to passivate, with suitable chemistry during these processes is of major interest.

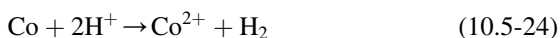
Although Co can be deposited electrochemically, one of the demerits of this process is hydrogen evolution. Thus unlike Cu ECD, Co cannot be deposited with 100% current efficiency (Faradaic efficiency) because a portion

of the applied current is consumed by the hydrogen evolution reaction. Owing to H_2 evolution, $Co(OH)_2$ precipitates at the metal surface. This current efficiency can be improved by optimizing various electroplating parameters of the electrolyte bath, such as pH, temperature, current density at which the deposition occurs, and agitation of the bath [221,222]. However, conformal deposition is achievable via the bottom fill approach for Co ECD when using suitable additives. As in Cu ECD, various suppressors and accelerators are suggested as additives for Co ECD. For example, 2-mercapto-5-benzimidazole sulfonic acid and thiourea are used as suppressor and accelerator, respectively, for Co ECD [223].

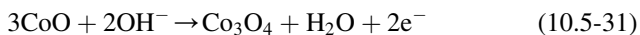
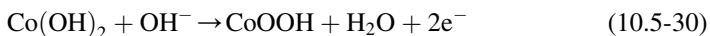
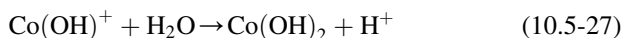
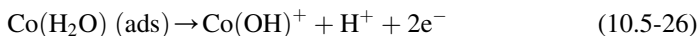
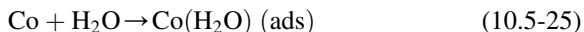
The deposition of Co by CVD and ALD is currently being explored and is preferred for smaller technology nodes [224–228]. Various precursors are claimed for both CVD and ALD processes. For example, the precursors such as Cp_2Co (where $Cp = C_5H_5$, cyclopentane), $CpCo(CO)_2$, and $Co(Co)NO$. $Co_2(CO)_8$ are suggested to deposit Co films using CVD [223–228]. Owing to the decomposition of metal halides at high temperature ($>500^\circ C$), metal organic precursors are preferred because deposition can be carried out at lower temperatures. Similarly, organometallic reagents with reducing agents such as H_2 or NH_3 have been proposed for ALD processes [229–236]. However, for ALD processes, a higher temperature is required to achieve sufficient growth rate. Unfortunately, a high temperature initiates undesired film growth via a parallel CVD process. Plasma and thermal ALD processes also are explored but have limitations such as poor conformal deposition in high-aspect ratio structures [237]. The direct liquid evaporation–CVD method is suggested to achieve a sufficient growth rate [239]. Obtaining C-, N-, and O-free Co film is one of the main requirements of the Co deposition process. It is found that impurity-free Co film is obtained when NH_3 is used along with H_2 as the coreactant [238]. Using only one of the reactants can result in the incorporation of C and N in the Co film. The ratio of the two reactants plays a significant role in determining the crystal structure of the Co and its resistivity [238]. The crystal structure obtained is a mixture of face-centered cubic (FCC) and hexagonal closed pack–Co. Post-CVD rapid thermal annealing (RTA) is carried out to increase the grain size, which decreases the resistivity. Also, the FCC crystallinity increases during RTA [239].

10.5.2 Passivation Process and Contamination Issues

After deposition, Co must be polished by CMP for delineation. However, Co is susceptible to corrosion via the dissolution of Co^{2+} ions as shown in Eq. (10.5-24). Thus tuning the slurry chemistry with suitable oxidizers and inhibitors is critical to passivate the Co surface and to control the polish rate and the defects.



Cobalt forms two types of passive layers: (1) Co(II) oxide/hydroxides, which usually form at lower potential, and (2) Co(III) oxide, which forms at higher potentials because of the oxidation of Co(II) oxides/hydroxides. The formation of various forms of oxides/hydroxides in different media at different pH values was investigated and reported in the literature [240,241]. Examples of the anodic reactions are given in Eqs. 10.5-25–10.5-28 [240].



The Pourbaix diagram shown in Fig. 10.5-15 reveals that self-passivation occurs only at neutral and alkaline media, whereas in acidic media active dissolution occurs. In neutral and alkaline media, the passive film mainly consists of CoO; Co(OH)₂ is present in the inner layer, whereas at the outer layer, the Co(III) oxides (Co₂O₃ and CoOOH) are present because of further oxidation of Co(OH)₂ [241].

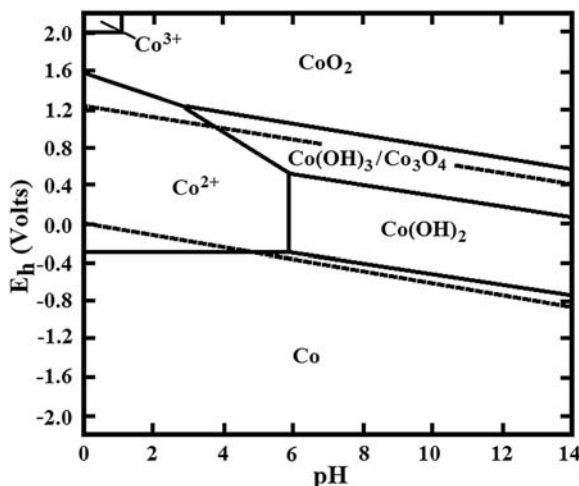


FIGURE 10.5-15 Pourbaix diagram for Co-H₂O system at 25°C [9,40]. Copyright National Association of Corrosion Engineers (NACE) International 1974. Used with permission.

During the Co CMP process, the formation of a passive film and its removal are dynamic in nature. By controlling the concentration of abrasives, oxidizers, and inhibitors, the removal rate and defects can be controlled. Various oxidizers and inhibitors are used for Co CMP processes; similar to the Cu CMP process, H_2O_2 and BTA have been studied as oxidizer and inhibitor, respectively. Lu et al. [242] reported that the adsorption of BTA on the Co surface occurs via both physisorption and chemisorption, with the adsorption isotherm following Langmuir-type adsorption. Peethala et al. [243] reported that the passive films mainly consist of Co(III) oxides in an arginine- H_2O_2 -based system and the addition of BTA to the slurry further decreases the removal rate. Brusic et al. [244] found that addition of Cu^{2+} ions to the Co slurry forms a thin film of Cu-BTA complex on the Co surface. The Cu-BTA acts as a better inhibitor especially in neutral media than the BTA alone.

Han et al. [9] proposed the mechanism for the Co CMP process. According to this mechanism, defects can be controlled in addition to controlling the removal rate if the slurry chemistry is judiciously tuned. To achieve this, the concentration of oxidizer, accelerator, and inhibitor must be optimized.

BTA is widely used as a Cu corrosion inhibitor in the semiconductor industry. However, it has issues with its use, such as attachment of abrasive particles and organic residues, both of which are also encountered during use in the Co post-CMP cleaning process. This is because both Co-BTA and Co oxide surfaces are hydrophobic in nature, facilitating the chemical and physical phenomena of particle and residue attachment to the surface. Thus it is desirable for post-CMP cleaning to be carried out with a suitable cleaning solution that renders the Co surface hydrophilic.

The main issues with Co post-CMP cleaning processes are uniform corrosion from chemical contact and galvanic corrosion between Co and Cu that are in contact with each other. Thus cleaning solutions must be formulated in such a way that they reduce both uniform and galvanic corrosion. The main driving force behind galvanic corrosion is the open circuit potential (OCP) difference between Co and Cu. Thus reducing the OCP difference between Co and Cu by controlling the cleaning chemistry is one approach. Alkaline solutions are preferred because the dissolution rate is lower as a result of the formation of a thin $\text{Co}(\text{OH})_2$ layer. A Co cleaning solution containing glycolic acid and BTA was studied [245]; however, the surface becomes hydrophobic with BTA adsorption, which is not preferred. Zhong et al. [246] found from the potentiodynamic polarization studies that the addition of 1,2,4-triazole (TAZ) to citric acid solutions reduces the galvanic corrosion and also makes the surface hydrophilic. They also found that the adsorption of TAZ is more effective on $\text{Co}(\text{OH})_2$ than on Co surfaces and the adsorption follows Langmuir adsorption isotherm.

10.6 SUMMARY

In this chapter, the surface morphology, deposition, and passivation mechanism of conducting materials such as Al, Cu, W, and Co during deposition, polishing, and cleaning are reported. Aluminum is a superior conducting material, which has been used in the IC industry for six decades. However, for many device types, it has been replaced by Cu because of its superior electrical properties, such as higher electrical conductivity and lower electromigration resistance. Similarly, W is used as via and contact conducting material for ICs. As the IC pattern size is reduced from micrometers to nanometers, the transition metal with superior properties such as Co may replace the traditional Cu and W. Among the various deposition techniques, Al is usually deposited by sputtering, Cu is deposited by ECD, W is deposited by PVD and CVD, and Co is deposited by PVD, CVD, and ECD. Although PVD and CVD are more commonly employed for Co, ECD may be the optimal choice because of its lower cost. During these deposition techniques, the processing conditions must be optimized to obtain a stable microstructure and optimum texture of the deposited metal. Postdeposition processing, such as annealing, is carried out to increase the grain size of the metal deposits. The surface morphology of these deposited films is reported. Chemical mechanical planarization is the technology used to planarize metals to remove the overburden material of the film surface. The final microstructure and texture obtained in the deposition techniques influence the CMP performance, as discussed in this chapter. Passivation of the metal surfaces during CMP is of utmost importance to control the polishing rate and surface defects. This chapter reports the various passivating agents used in wet chemistry, including CMP slurry for the different metal surfaces. The post-CMP step is used to remove the organic residues and metal contaminants by cleaning the substrate with several chelating and cleaning solutions, which are generated from CMP. This chapter also highlights and suggests the mechanism of removal of defects generated during CMP and post-CMP of metals.

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